

DealLens

AI-powered PE Deal Screening & Underwriting Platform

From company discovery to investment committee-ready underwriting.

Core design principle

Deterministic finance engine + auditable AI layer

Prepared as a build blueprint for Claude Code / Codex



1. Product thesis

What the product should accomplish and why the workflow matters.

PRODUCT OBJECTIVE

Goal

Turn an initial company idea into a structured PE underwriting case with minimal manual work.

The platform does not replace analyst judgment. It compresses repetitive research, standardizes assumptions, and makes the investment logic auditable.

- Standardize first-pass screening across many targets.
- Keep financial calculations deterministic and reproducible.
- Use AI for extraction, classification, synthesis, and challenge - not as the source of truth for core math.
- Generate outputs that look and read like an internal investment work product.

Success criteria

TARGET TIME

< 10 min

AUDITABILITY

100%

CORE OUTPUTS

8+

SCENARIO CASES

3

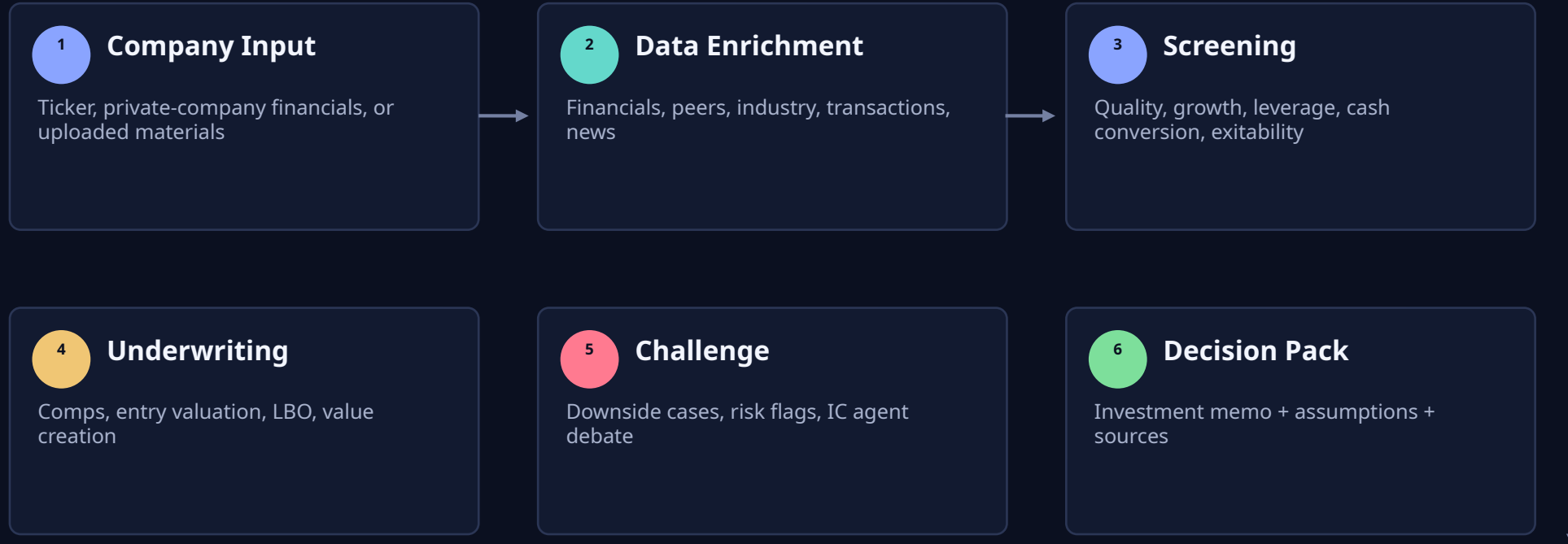
Outputs

Screening score | comps | entry valuation | LBO | sensitivities | risks | DD questions | IC memo

2. End-to-end workflow

The product should feel like one continuous investment process, not separate mini-apps.

USER JOURNEY



Core principle

Every major output should link back to an input, formula, source document, or explicit AI reasoning trace.

3. System architecture

Recommended stack for a serious portfolio-grade implementation.

ARCHITECTURE



Deployment Vercel / Railway or Render / Supabase / GitHub Actions

4. Data model

Design the database around investment objects, not screens.

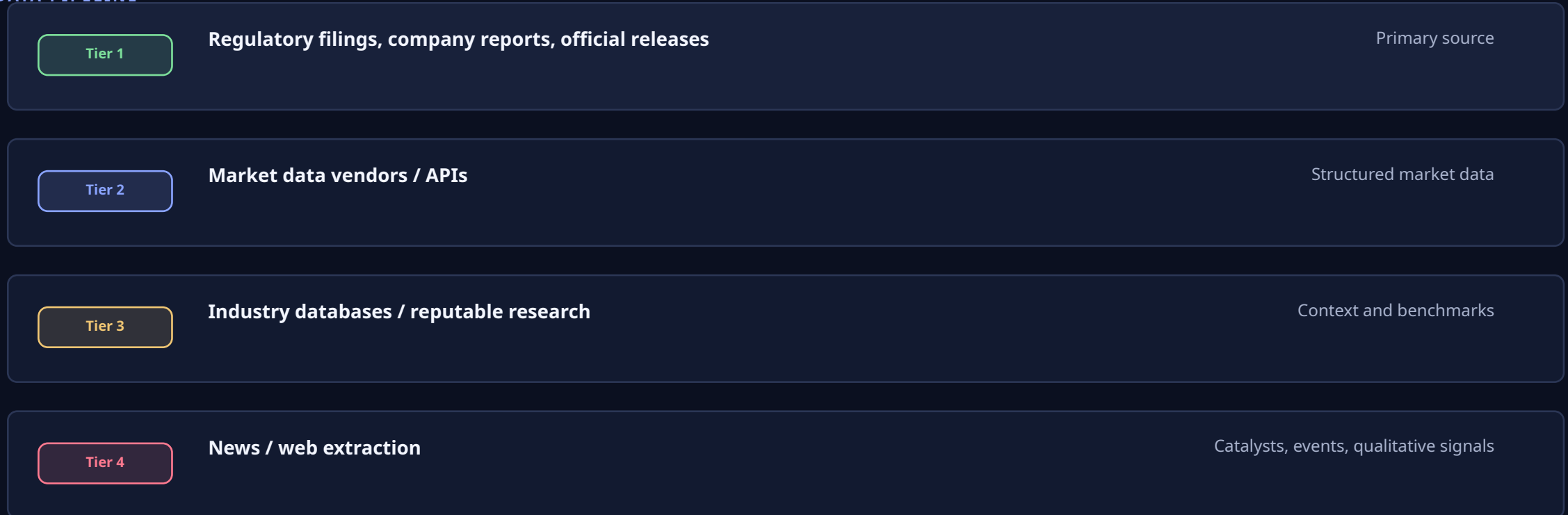
CORE ENTITIES

Company identity, sector, geography, ownership	FinancialPeriod revenue, EBITDA, cash flow, debt, capex
Peer peer identity + similarity attributes	Transaction deal value, multiples, buyer, target, date
Assumption name, value, unit, source, confidence	Scenario base / bull / bear + overrides
LBOCase sources, uses, debt, returns, sensitivities	Memo sections, citations, version, timestamp
Key rule Store raw data, normalized data, assumptions, and AI outputs separately so the model is reproducible.	

5. Data ingestion & source hierarchy

Source quality determines model credibility.

DATA PIPELINE



Normalization

Currency -> EUR base | accounting periods -> canonical calendar | units -> millions | signs -> consistent | source timestamp preserved

6. PE attractiveness engine

A transparent scorecard, not a black-box AI rating.

SCREENING

Business quality Recurring revenue, retention, market position	20%	Growth Revenue CAGR, organic growth, TAM	15%
Margins EBITDA margin, stability, operating leverage	15%	Cash conversion FCF / EBITDA, capex intensity, NWC	15%
Leverage capacity Debt serviceability, cyclicalicity, covenant headroom	10%	Market structure Fragmentation, pricing power, competition	10%
Exit optionality Strategic buyers, sponsor appetite, market depth	10%	Management / execution Track record, key-person risk, governance	5%

Score formula

Score = $\sum(\text{weight}_i \times \text{normalized factor}_i)$, with factor definitions stored in the database and versioned.

7. Financial quality module

Convert raw financials into PE-relevant operating diagnostics.

FINANCIAL ANALYSIS

Growth

Revenue CAGR
Organic vs inorganic
Bookings / ARR where relevant

Profitability

Gross margin
EBITDA margin
Margin trajectory

Cash flow

FCF conversion
Capex intensity
NWC drag / release

Balance sheet

Net debt / EBITDA
Interest coverage
Liquidity headroom

Diagnostic outputs

Margin expansion potential

Cash conversion quality

Leverage headroom

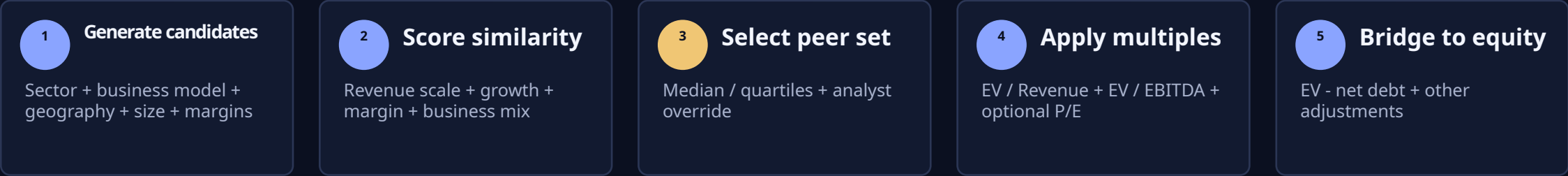
Cyclicality flag

Each diagnostic should have a simple rule, a data source, and an explanation field.

8. Comparable companies engine

Automate peer discovery, then preserve analyst override.

VALUATION



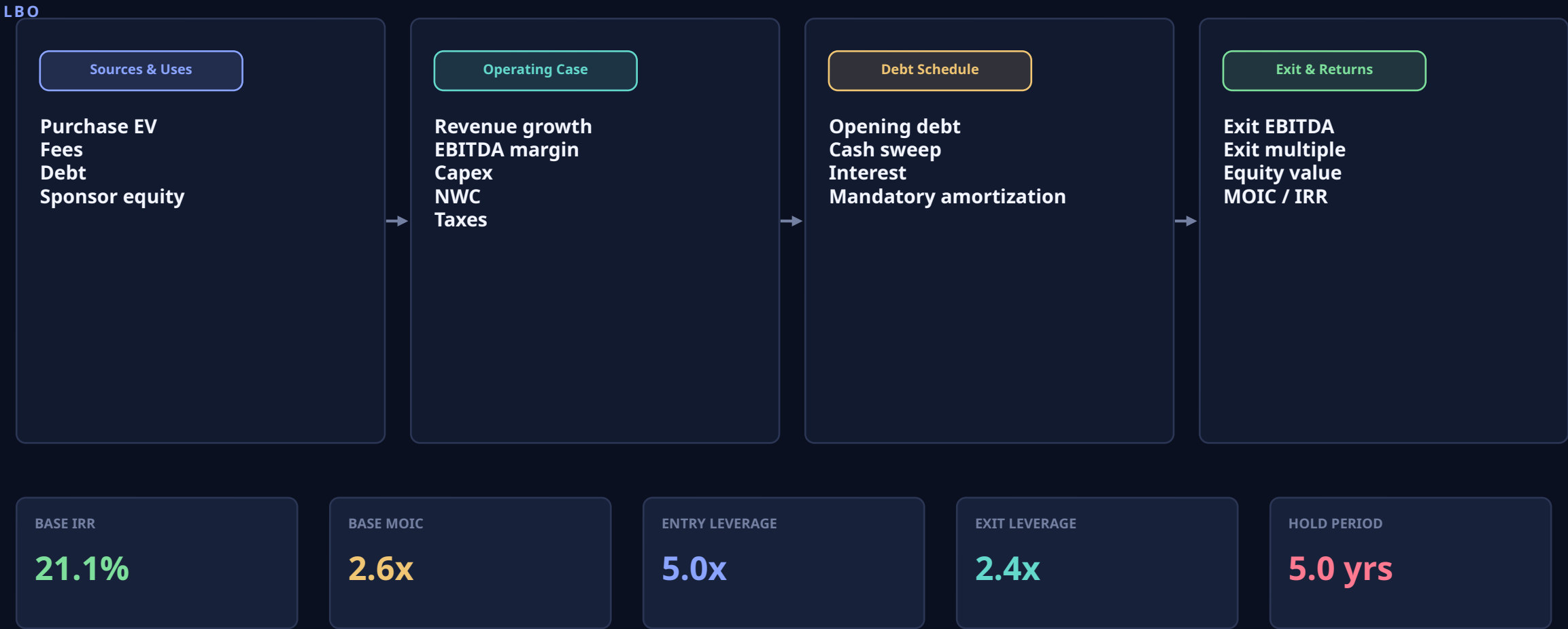
Illustrative output

Peer	EV/Revenue	EV/EBITDA	Margin	Similarity
Peer A	3.2x	12.4x	26%	91
Peer B	3.8x	13.1x	29%	87
Peer C	2.9x	11.8x	24%	82
Median	3.2x	12.4x	26%	-

Important: peer selection should never be silently accepted. Store selected peers + rejected candidates + reason codes.

9. LBO underwriting engine

The core deterministic finance model.

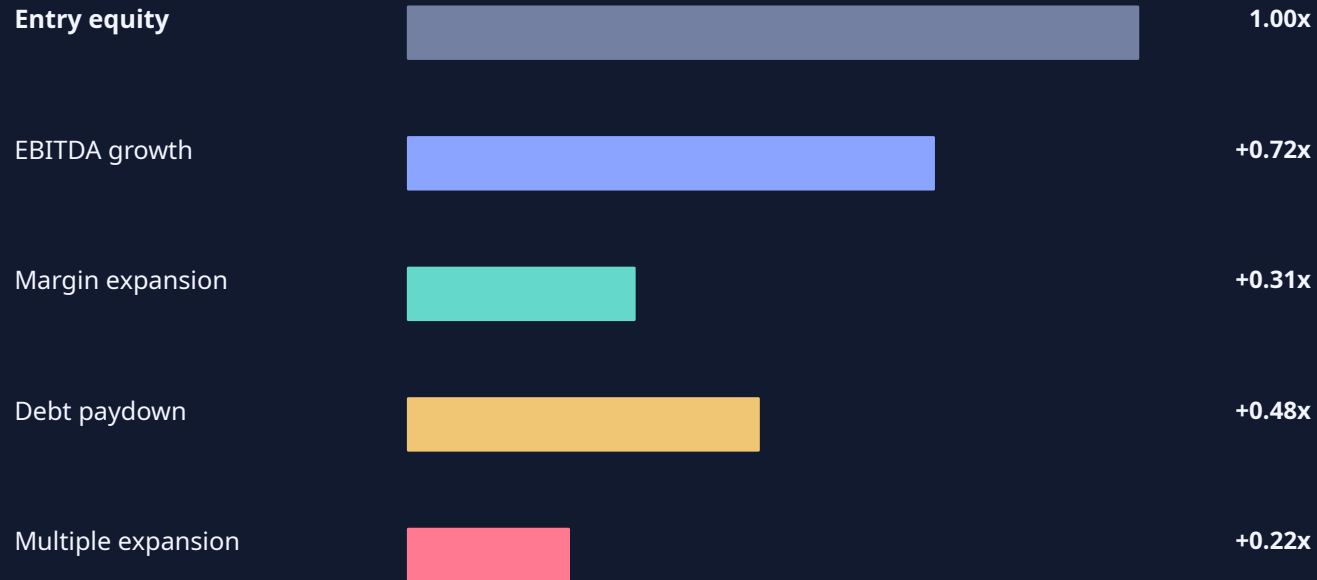


10. Value creation bridge

Explain what actually drives returns.

RETURNS ATTRIBUTION

MOIC attribution



Illustrative only - the calculation should reconcile exactly to the model outputs.

Investment implication

- Separates operational value creation from multiple expansion.
- Makes sponsor underwriting more explainable.
- Lets the IC test whether returns depend on aggressive assumptions.

Red flag

If >50% of value creation comes from multiple expansion, label the case as exit-multiple dependent.

11. Scenario & sensitivity framework

Every underwriting case should expose what can break the thesis.

Downside Analysis

BULL

High growth
margin expansion
flat-to-up exit multiple

BASE

Consensus / normalized
operating assumptions
market exit multiple

BEAR

Growth slowdown
margin compression
-2.0x exit multiple

Sensitivity matrix examples

Entry / Exit	10x	11x	12x	13x
10x	18%	21%	24%	27%
11x	15%	18%	21%	24%
12x	12%	15%	18%	21%
13x	9%	12%	15%	18%

Automate tornado views for: growth, margin, entry multiple, leverage, interest rate, exit multiple.

12. Risk engine & due diligence questions

Turn model outputs into a practical diligence agenda.

RISK & DILIGENCE

Customer concentration

Revenue exposure to top customers

FINANCIAL

Margin sustainability

Recent expansion may be cyclical

OPERATING

Leverage

Debt serviceability under bear case

FINANCING

Multiple risk

Returns rely on re-rating

VALUATION

Key-person dependency

Founder / management concentration

EXECUTION

Exit liquidity

Limited sponsor / strategic buyer universe

EXIT

AI-generated DD questions

Questions should be tied to specific model assumptions: churn, pricing, customer concentration, capex normalization, add-on pipeline, management incentives, refinancing risk.

13. AI layer - where Claude / Codex adds real leverage

AI should augment the investment process without contaminating the core model.

AI ARCHITECTURE

Document parser

Extract metrics, tables, guidance, risks, named entities

PDF / HTML / OCR

Research synthesizer

Summarize filings + news into structured evidence

Citations required

Peer classifier

Explain why a company is a comparable or not

Confidence + reason

Thesis builder

Convert model + research into investment thesis

Source-backed

IC challenger

Attack assumptions and surface contradictions

Bull / Bear / Risk agents

Memo writer

Generate standardized IC pack

Template + citations

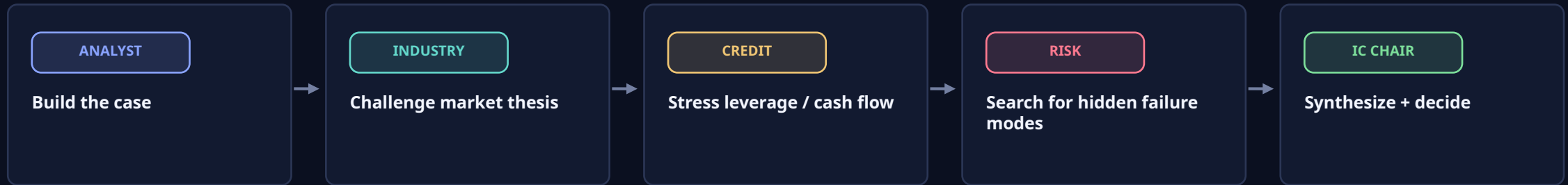
Guardrail

AI may propose or explain. Python owns the financial math. Every factual claim should retain a source or be labeled as an assumption.

14. Investment Committee simulator

A differentiated feature for the final underwriting step.

AGENT WORKFLOW



Expected output

Recommendation

PROCEED TO DD

Key strengths

Cash conversion | recurring revenue | fragmented market

Key risks

Entry valuation | concentration | refinancing

Unanswered DD

Retention | pricing power | management depth

Decision rule

Do not permit a positive recommendation if critical assumptions fail the bear case thresholds.

15. IC memo generator

Package the entire model into one decision-ready artifact.

DELIVERABLE

01 **Transaction overview**

02 **Company & industry**

03 **Investment thesis**

04 **Financial performance**

05 **Comparable valuation**

06 **LBO returns**

07 **Value creation plan**

08 **Risks & downside**

09 **DD questions**

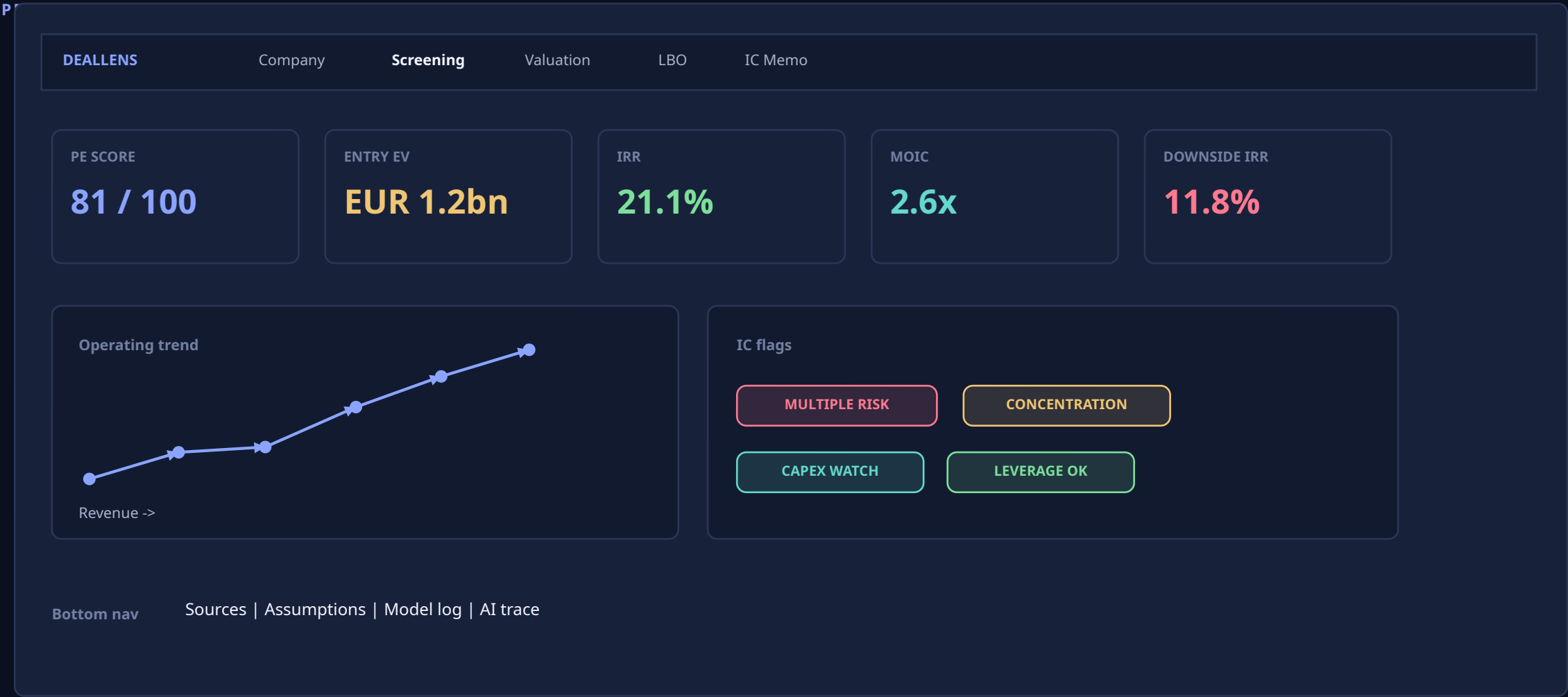
10 **Recommendation**

Versioning

Memo stores scenario, model version, source timestamps, selected peers, and key assumptions so the output can be reproduced later.

16. Dashboard UX

Keep the interface close to an analyst workstation.



17. Step-by-step build plan

Build the deterministic core first, then layer AI and polish.

IMPLEMENTATION ROADMAP

Phase 0	Repo + schema	Monorepo, DB schema, environment, CI, seed data
Phase 1	Data layer	Company financials, peers, sources, ingestion jobs
Phase 2	Finance engine	Screening score, comps, valuation, LBO, scenarios
Phase 3	UI	Dashboard, company page, model pages, memo viewer
Phase 4	AI layer	Document parser, synthesis, IC agents, citations
Phase 5	QA + demo	Unit tests, reconciliation, edge cases, deployment

Priority order **Correctness -> auditability -> workflow -> AI -> visual polish**

18. How to use Claude Code / Codex

Use the coding agent as an orchestrator with explicit acceptance criteria.

AGENTIC DEVELOPMENT

Build one vertical slice

"Implement Company -> Financials -> Screening Score end-to-end. Use typed schemas, tests, and seed data. Do not build LBO yet."

Add deterministic module

"Implement LBO engine with explicit inputs, formula tests, and reconciliation checks. Return structured JSON plus traceable assumptions."

Add AI safely

"Implement filing extraction with source spans, confidence, and a human-review queue. Never write extracted values directly into the financial model without validation."

Harden the app

"Add tests for missing data, currency mismatch, negative EBITDA, debt covenant breach, null peers, duplicate sources, and scenario overrides."

Workflow rule

Always ask the coding agent to show file changes, run tests, and report unresolved assumptions before moving to the next feature.

19. Testing & model validation

A finance product is only credible if the numbers reconcile.

QUALITY CONTROL

Unit tests

LBO formulas, IRR, MOIC, debt schedule, sensitivities

Reconciliation tests

Sources = Uses; opening debt + changes = closing debt

Scenario tests

Bull / base / bear override logic behaves predictably

Data tests

Currency, units, periods, missing values, duplicate sources

AI tests

Citation presence, extraction accuracy, hallucination checks

UI tests

Critical workflows from company input to memo export

Definition of done

A demo company can be loaded, every number traces to a formula/source, scenarios reconcile, AI claims cite evidence, and the IC memo is reproducible.

20. MVP scope vs v1.0

Protect the project from turning into an endless platform build.

SCOPE CONTROL

MVP

- 1-2 public data sources
- Company financial profile
- PE attractiveness score
- Basic peer comps
- Entry valuation
- 5-year LBO
- Base / bull / bear
- 2-page memo export
- Core unit tests

V1.0

- Broader source connectors
- Private-company upload workflow
- Advanced peer classifier
- Precedent transactions
- Debt sculpting / multiple tranches
- Tornado + scenario engine
- Multi-agent IC debate
- Memo citations + source map
- Cloud deployment + auth

21. How to position it on LinkedIn

The project should demonstrate investment judgment, not just coding.

PUBLIC POSITIONING

Post 1

The problem

"I wanted to recreate the first-pass PE underwriting workflow in software."

Post 2

The numbers

Show a real company case: score, comps, LBO, downside.

Post 3

The insight

Show what actually drives the return: growth, deleveraging, margins, multiple.

Post 4

The architecture

Explain why financial math is deterministic and AI is an evidence layer.

Post 5

The release

GitHub + short demo + what you learned building it.

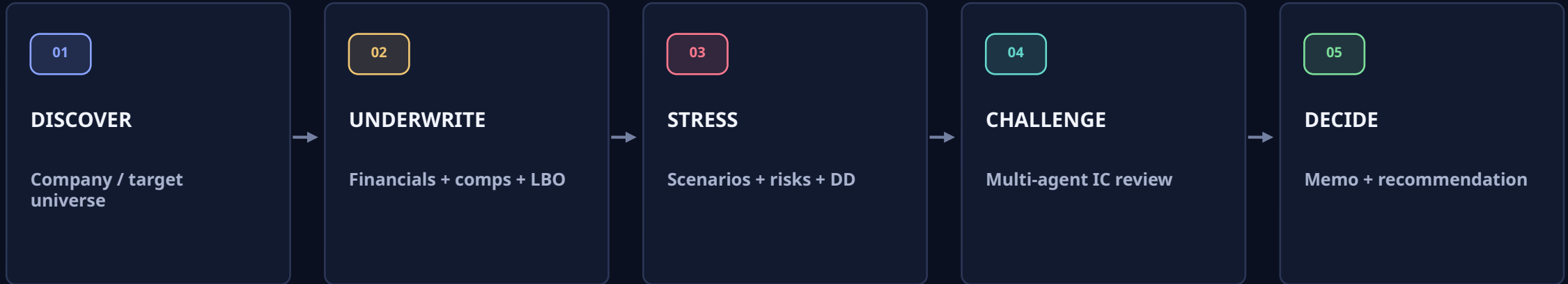
Positioning line

"I built an AI-assisted PE underwriting platform that takes a company from screening to an IC-ready investment case."

22. Final blueprint

What the finished project should look like.

TARGET STATE



End state

A credible, demoable investment workflow that combines finance fundamentals, data engineering, and AI - with explicit auditability.

The strongest version is not the one with the most features. It is the one where every output is economically interpretable and technically reproducible.